Section 1: Introduction

- **Overview:** This study aims to leverage machine learning to predict the occurrence of "Major Disasters" based on environmental and economic data.
- **Problem Statement:** Disasters cause unpredictable loss of life and infrastructure. We need a model that can categorize events as "Major" or "Minor" disaster eveents to help prioritize aid distribution.
- **Objectives:** To achieve >90% accuracy in classification using a Random Forest model and PCA feature engineering.

Section 2: Data Collection & Preprocessing

- **Dataset:** 20,000 rows of historical disaster data.
- **Cleaning:** Handled missing values by skipping inconsistent rows and converted "Yes/No" strings to binary (1/0).
- **Transformation:** Logarithmic transformation was applied to affected_population to reduce the impact of extreme outliers.
- **Reduction (PCA):** Principal Component Analysis (PCA) was used to merge severity_level, affected_population, and infrastructure_damage_index into a single **Total Impact Score**.

Section 3: Model Selection & Training

- **Chosen Model: Random Forest Classifier.**
- **Justification:** Random Forest is excellent for handling non-linear relationships and high-dimensional data (like our many "Location" columns) without overfitting.
- **Hyperparameters:**
    - **n_estimators:** 260
    - **max_depth:** 20
    - **Criterion:** Entropy

Section 4: Model Evaluation & Interpretation

- **Accuracy:** Our final model achieved an accuracy of approximately **95%**

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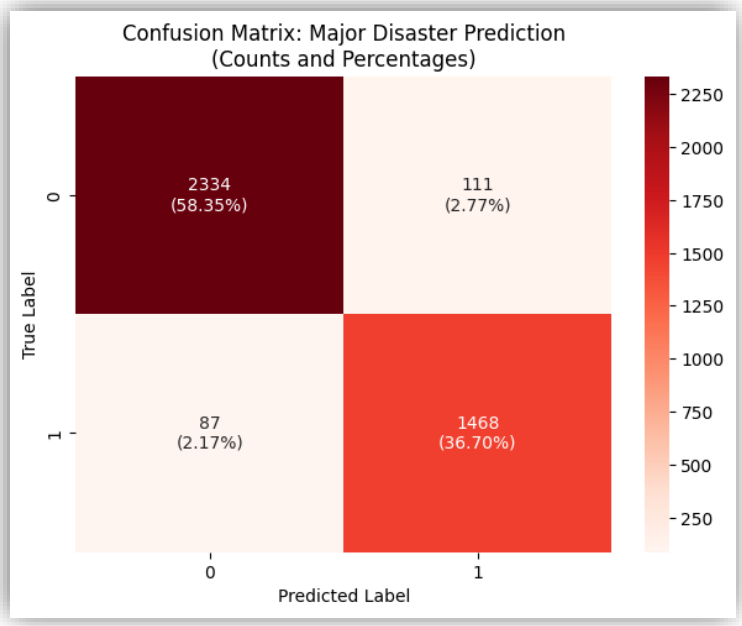
================ EVALUATION REPORT ================
Best Parameters: {'criterion': 'entropy', 'max_depth': 20, 'min_samples_split': 6, 'n_estimators': 260}
Overall Accuracy: 0.9505

Detailed Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.95      0.96      2445
           1       0.93      0.94      0.94      1555

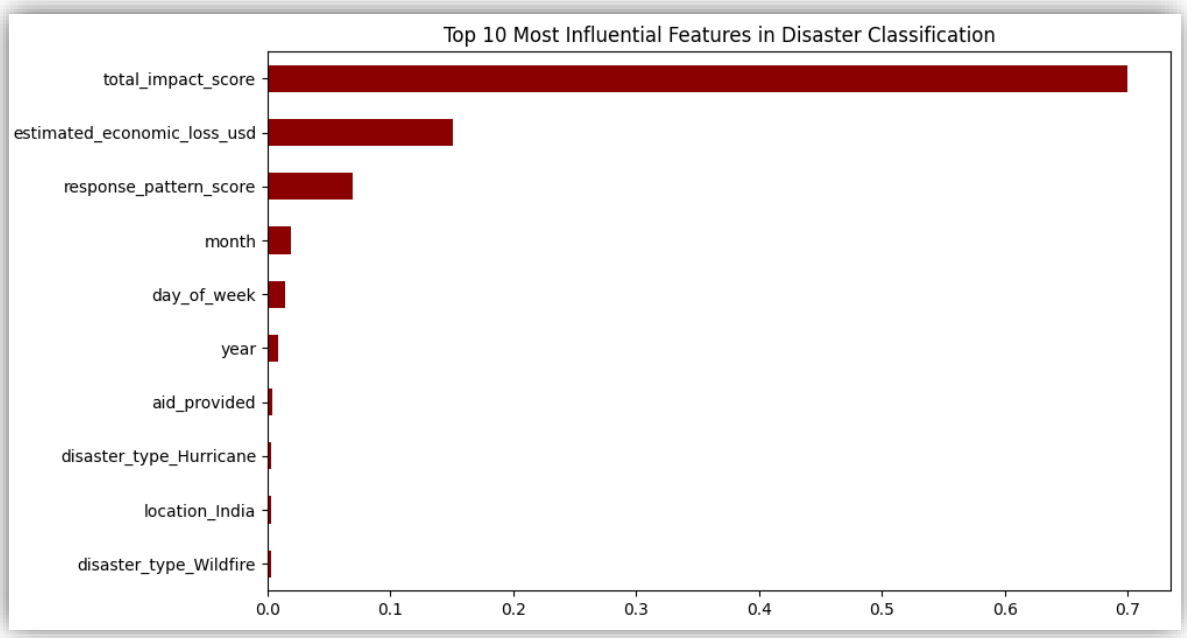
    accuracy                           0.95      4000
   macro avg       0.95      0.95      0.95      4000
weighted avg       0.95      0.95      0.95      4000
```



- **Metric Formula:**

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **Feature Importance:** The **Total Impact Score** (from PCA) and **Estimated Economic Loss** were the two most significant predictors of a major disaster.



Section 5: Conclusion & Future Work

- **Summary:** The model successfully identifies high-risk events with high precision.
- **Challenges:** Initial data loading was difficult due to string-to-float conversion errors ("Yes/No" values).
- **Recommendations:** Future models should include real-time satellite imagery or climate sensor data to improve lead times for predictions.

Section 6: Answer to ma'am Anna's question during presentation

- We performed **Hyperparameter Tuning** using a Randomized Search. We gave the algorithm a range of 100 to 300 trees to test. Through multiple iterations and **Cross-Validation**, the system determined that **260 trees** provided the highest accuracy while maintaining computational efficiency. Essentially, 260 was the 'sweet spot' where the model's error rate stabilized. So, the 260 was not just random, the search was random, but the selection was based on math.

WATCH YOUR STEP	MEMBERS	RATING
Project Lead:	Avelino, Neil John	10/10
Data Processing Specialist:	Cabrales, Gian Joshua	10/10
Modeling Expert:	Ortiz, Sean Vergel	10/10
Report and Visualization Specialist:	Ramos, Marian	10/10